

Hybrid Quantum 4-Sieve

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Abstract

We present an adaptation of the quantum random walk (QRW) 2-sieve to serve as the inner two-list subroutine inside a quantum 4-sieve. Inspired by the classical 4-sieve, we construct the quantum 4-sieve and use the QRW 2-sieve as its inner two-list subroutine. The adaptation preserves the Johnson graph, the second-level random product code, and the quantum data structure of the original algorithm, and modifies only the marked condition and the probability that a random cross-pair is a solution. We provide the complete algorithms, discuss the modifications required, and give a full complexity analysis for the running time, QRAM, quantum memory, and classical memory. We optimise the QRW parameters for the first two inner calls, and also analyse the third inner call in detail...

1 Introduction

The Shortest Vector Problem (SVP) is at the foundation of lattice-based cryptography. The best known heuristic quantum algorithm for SVP runs in time $2^{0.2570d+o(d)}$ [1], obtained by replacing Grover's algorithm inside the 2-sieve by a quantum random walk with an additional layer of locality-sensitive filtering. In parallel, k -sieve algorithms have been developed to reduce memory requirements at the cost of increased running time [2]. In particular, the quantum 4-sieve of [2] uses nested Grover searches and achieves a time-memory trade-off with quantum-accessible classical memory of size $2^{0.1724d}$ to $2^{0.1813d}$.

In this work, we investigate whether the QRW 2-sieve of [1] can be used as the inner two-list subroutine inside the 4-sieve framework of [2]. This requires adapting the QRW 2-sieve to handle two distinct lists rather than one, an arbitrary configuration threshold $C'_{12}(\alpha)$ rather than the fixed angle θ_α^* , and possibly non-unit vectors in the second-level call.

We show that the core QRW structure remains unchanged: the Johnson graph, the second random product code $\mathcal{C}_2$, the quantum data structure for the lists $J^v(\vec{t})$, and the MNRS framework are all preserved. Only the marked condition and the probability p_{sol} that a random cross-pair is a solution must be adjusted. We give the complete algorithms, discuss the modifications, and provide the full complexity analysis.

2 Preliminaries and Notation

We use the notation of [1, 2]. Let d be the lattice dimension and define the fundamental list-size parameter

$$N := 2^{cd+o(d)}. \quad (1)$$

This quantity N is exponential in d and serves as the reference scale for all exponents in the algorithm. A list of size N^a contains $2^{a \cdot cd+o(d)}$ elements; throughout the paper we write $N^{a+o(1)}$ to denote such a list, absorbing subexponential factors into the $o(1)$. For an angle $\alpha \in [0, \pi/2]$, the volume ratio of a spherical cap of angle α is

$$\mathcal{V}_d(\alpha) := \text{poly}(d) \sin^d(\alpha), \quad (2)$$

which represents the fraction of the unit sphere covered by a cap of angular radius α . For two angles α and θ , the volume ratio of a wedge formed by two caps of angle α with centers separated by angle θ is

$$\mathcal{W}_d(\alpha, \theta) := \text{poly}(d) \left(1 - \frac{2 \cos^2(\alpha)}{1 + \cos(\theta)} \right)^{d/2}, \quad (3)$$

which is well-defined provided $2 \cos^2(\alpha) < 1 + \cos(\theta)$, i.e., the two caps overlap. The critical angle between residual vectors that characterises reducibility of two border points is

$$\theta_\alpha^* := 2 \arcsin \left(\frac{1}{2 \sin(\alpha)} \right), \quad (4)$$

as established in [1, Proposition 4]. Two residual vectors $\vec{y}_0, \vec{y}_1$ on the border of a filter of angle α are reducible if and only if $\theta(\vec{y}_0, \vec{y}_1) \leq \theta_\alpha^*$.

Outer 4-sieve parameters. The outer 4-sieve uses a prefiltering angle $\alpha_0 \in [\pi/3, \pi/2]$ and a configuration matrix $C \in \mathbb{R}^{4 \times 4}$. The reduction to residual vectors is denoted $C'(\alpha_0)$, with entries $C'_{ij}(\alpha_0)$ given by [2, Lemma 1]:

$$C'_{ij}(\alpha_0) = \frac{1}{\sin^2(\alpha_0)} \left(C_{ij} + \frac{\cos^2(\alpha_0)}{3} \right) \quad (i \neq j). \quad (5)$$

The effective threshold for the second-level 2-sieve that builds L_{1234} is

$$C_{\text{eff}} := \frac{\sin^2(\alpha_0)}{2R^2} - 1, \quad R^2 := 2 + 2C_{12}, \quad (6)$$

as given in [2, Lemma 6].

QRW parameters. The adapted QRW 2-sieve uses four free parameters c, c_1, c_2, ρ satisfying $0 \leq c_2 \leq c_1 \leq c \leq 1$ and $0 \leq \rho \leq \rho_0$. From these, the following derived quantities are defined. The angle α (not to be confused with α_0) satisfies $\mathcal{V}_d(\alpha) = N^{-(1-c)}$. The angle β of the second random product code satisfies $\mathcal{V}_d(\beta) = N^{c_2-c_1}$. The target angle is $\theta' := \arccos(C')$. The collision parameter ρ_0 is defined by $N^{\rho_0} := \mathcal{V}_d(\beta)/\mathcal{W}_d(\beta, \theta')$. The solution probability is

$$p_{\text{sol}} := \frac{1}{2} (1 - (C')^2)^{d/2}, \quad (7)$$

where the factor $1/2$ accounts for the bipartite nature of the problem. The expected number of solutions per filter is $N^\zeta := N^{2c} \cdot p_{\text{sol}}$.

The QRW costs are the setup cost $\mathcal{S} := N^{c_1+\rho}$, the update cost $\mathcal{U} := N^{\max\{\rho, (\rho+c_2)/2\}}$, and the checking cost $\mathcal{C} := 1$. The spectral gap of the Johnson graph is $\delta := N^{-c_1}$. The fraction of marked vertices is

$$\epsilon := N^{2c_1} \cdot p_{\text{sol}} \cdot N^{\rho-\rho_0}. \quad (8)$$

3 Modified QRW 2-Sieve

3.1 Preserved components

The adaptation preserves the following components of the QRW 2-sieve of [1] without structural modification. The Johnson graph $J(N, N^{c_1})$ on the combined list, with spectral gap $\delta = N^{-c_1}$ and the MNRS quantum walk framework, is retained exactly. The second random product code $\mathcal{C}_2$ with angle β satisfying $\mathcal{V}_d(\beta) = N^{c_2-c_1}$ and size $|\mathcal{C}_2| = N^{\rho+c_1-c_2}$ is constructed identically. The quantum data structure for the lists $J^v(\vec{t}) = f_\beta(\vec{t}) \cap L^v$, supporting insertions and deletions in quantum superposition, is used as in the original algorithm. The setup, update, and check cost formulas are formally unchanged.

3.2 Required modifications

Three modifications are required to adapt the QRW 2-sieve to the bipartite setting. First, we form the combined list $L = R_A \cup R_B$ and tag each element by a bit indicating its origin. The Johnson graph is built on L with $|L| = 2N_R$. This tagging does not affect the graph structure. Second, the marked condition is redefined: a vertex v is marked if there exist $\vec{t} \in \mathcal{C}_2$ and distinct $\vec{y}_i, \vec{y}_j \in \tilde{J}^v(\vec{t})$ such that $\vec{y}_i$ and $\vec{y}_j$ have different tags and $\langle \vec{y}_i | \vec{y}_j \rangle \leq C'$. Third, the solution probability acquires the factor $1/2$ as given in Equation (7).

3.3 Adapted algorithm

Algorithm 1 Adapted QRW 2-Sieve for bipartite configuration problems

Require: two residual lists R_A, R_B of size N_R each; threshold C' ; QRW parameters

c, c_1, c_2, ρ .

Ensure: all pairs $(\vec{y}_a, \vec{y}_b) \in R_A \times R_B$ with $\langle \vec{y}_a | \vec{y}_b \rangle \leq C'$.

- 1: Form $L := R_A \cup R_B$ with tags.
 - 2: Compute $\theta' := \arccos(C')$ and $p_{\text{sol}} := \frac{1}{2}(1 - C'^2)^{d/2}$.
 - 3: Sample random product code $\mathcal{C}_2$ with $|\mathcal{C}_2| = N^{\rho+c_1-c_2}$ and angle β satisfying $\mathcal{V}_d(\beta) = N^{c_2-c_1}$.
 - 4: Build Johnson graph $J(N, N^{c_1})$ on L .
 - 5: **for** each vertex v **do**
 - 6: Compute $J^v(\vec{t}) = f_\beta(\vec{t}) \cap L^v$ for each $\vec{t} \in \mathcal{C}_2$.
 - 7: Truncate to $\tilde{J}^v(\vec{t})$ of size $\leq 2N^{c_2}$.
 - 8: Set marked bit $M(v)$ according to the marked condition above.
 - 9: **end for**
 - 10: **while** total solutions found $< N^\zeta/2$ **do**
 - 11: Pick a new random product code $\mathcal{C}_2$.
 - 12: **while** solutions found with this $\mathcal{C}_2 < N^{\zeta+\rho-\rho_0}/2$ **do**
 - 13: Run the MNRS quantum walk to find one marked vertex.
 - 14: **end while**
 - 15: **end while**
 - 16: **return** all collected pairs.
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4 Hybrid Quantum 4-sieve

Algorithm 2 Hybrid Quantum 4-sieve

Require: list L of $N' = N^{1+o(1)}$ lattice vectors of norm 1; reducing factor $\gamma < 1$; prefiltering angle α_0 ; configuration $C \in \mathbb{R}^{4 \times 4}$; QRW parameters.

Ensure: list L' of N' lattice vectors of norm $\leq \gamma$.

- 1: $L' := \emptyset$.
 - 2: **while** $|L'| \leq N'$ **do**
 - 3: Sample a 4-RPC $\mathfrak{C}$ with codewords $(\mathbf{A}_1^j, \mathbf{A}_2^j, \mathbf{A}_3^j, \mathbf{A}_4^j)$ such that $\sum_i \mathbf{A}_i^j = \vec{0}$ and $\langle \mathbf{A}_i^j | \mathbf{A}_{i'}^j \rangle = -1/3$ for $i \neq i'$.
 - 4: Partition L into tuple-filters: for each $\vec{x} \in L$, find its nearest codeword $\mathbf{A}_i^j$, add $\vec{x}$ to L_i^j , compute residual $\vec{y} = (\vec{x} - \cos \alpha_0 \mathbf{A}_i^j) / \sin \alpha_0$, and add $\vec{y}$ to R_i^j .
 - 5: **for** each tuple-filter j **do**
 - 6: $L_{12}^j \leftarrow \text{QRW-2-SIEVE}(R_1^j, R_2^j, C'_{12}(\alpha_0))$.
 - 7: $L_{34}^j \leftarrow \text{QRW-2-SIEVE}(R_3^j, R_4^j, C'_{34}(\alpha_0))$.
 - 8: $L_{1234}^j \leftarrow \text{GROVER}(\tilde{L}_{12}^j, \tilde{L}_{34}^j, C_{\text{eff}})$.
 - 9: $L' := L' \cup L_{1234}^j$.
 - 10: **end for**
 - 11: **end while**
 - 12: **return** L' .
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5 Complexity Analysis

We now give the full complexity of the adapted QRW 2-sieve and of the outer 4-sieve. All calculations are performed in the QRAM model and expressed in the exponent notation $N^{cd+o(1)}$.

5.1 Complexity of the adapted QRW 2-sieve

The setup cost, update cost, checking cost, and spectral gap are formally unchanged from the original algorithm:

$$\mathcal{S} = N^{c_1+\rho+o(1)}, \quad (9)$$

$$\mathcal{U} = N^{\max\{\rho, (\rho+c_2)/2\}+o(1)}, \quad (10)$$

$$\mathcal{C} = 1, \quad (11)$$

$$\delta = N^{-c_1}. \quad (12)$$

The fraction of marked vertices is modified by the factor $1/2$ in p_{sol} :

$$\epsilon = \frac{1}{2} N^{2c_1} (1 - (C')^2)^{d/2} N^{\rho-\rho_0}, \quad (13)$$

where ρ_0 satisfies $N^{\rho_0} = \mathcal{V}_d(\beta)/\mathcal{W}_d(\beta, \theta')$. The expected number of solutions per filter is

$$N^\zeta = \frac{1}{2} N^{2c} (1 - (C')^2)^{d/2}. \quad (14)$$

The time to find one marked vertex is

$$T_1 = \mathcal{S} + \frac{1}{\sqrt{\epsilon}} \left(\frac{1}{\sqrt{\delta}} \mathcal{U} + \mathcal{C} \right). \quad (15)$$

The time for the FINDALLSOLUTIONS subroutine is

$$\text{FAS}_1 = \max\{N^\zeta, 1\} \cdot T_1. \quad (16)$$

The total time of one QRW 2-sieve call is

$$T_{\text{QRW}} = \text{NbRep}_{\text{QRW}} \cdot (N + N^{1-c} \cdot \text{FAS}_1), \quad (17)$$

where

$$\text{NbRep}_{\text{QRW}} = \max\{1, N^{c-\zeta+o(1)}\}. \quad (18)$$

The derived parameters are:

$$\log_2 \sin(\alpha) = -(1-c) \cdot n, \quad (19)$$

$$\log_2 \sin(\beta) = (c_2 - c_1) \cdot n, \quad (20)$$

$$\theta' = \arccos(C'), \quad (21)$$

$$\rho_0 = \frac{\log_2 \mathcal{V}_d(\beta) - \log_2 \mathcal{W}_d(\beta, \theta')}{n d}, \quad (22)$$

$$\zeta = 2c + \frac{\log_2 p_{\text{sol}}}{n d}. \quad (23)$$

5.2 QRW 2-sieve on the standard 2-sieve problem

We first verify the framework by reproducing the result of [1, Theorem 1] for the standard 2-sieve problem. In this case, the list size is N (i.e., $n = 0.2075$), the threshold is θ_α^* with $p_{\text{sol}} = \mathcal{V}_d(\theta_\alpha^*)$, and the optimal parameters are

$$c \approx 0.3696, \quad c_1 \approx 0.2384, \quad c_2 = 0, \quad \rho \rightarrow 0. \quad (24)$$

With these values:

$$\log_2 \sin(\alpha) = -0.1308, \quad \alpha \approx 1.1514 \text{ rad}, \quad (25)$$

$$\log_2 \sin(\beta) = -0.0495, \quad \beta \approx 1.313 \text{ rad}, \quad (26)$$

$$\theta_\alpha^* \approx 1.158 \text{ rad}, \quad \rho_0 \approx 0.106, \quad \zeta \approx 0.132. \quad (27)$$

The QRW costs are:

$$\mathcal{S} = N^{0.2384}, \quad \mathcal{U} = N^{o(1)}, \quad \delta = N^{-0.2384}, \quad \epsilon = N^{-0.2364}. \quad (28)$$

Then $T_1 \approx N^{0.2384}$, $\text{FAS}_1 = N^{0.3704}$, and

$$T_{\text{QRW}} \approx N^{1.2384} = 2^{0.2570d+o(d)}, \quad (29)$$

which matches [1, Theorem 1].

5.3 QRW 2-sieve for the first two inner calls

We now apply the framework to the first two inner calls of the 4-sieve at the minimal-memory configuration of [2, Proposition 17]: $|L| = 2^{0.1724d}$, $\alpha_0 \approx 1.3131 \text{ rad}$, and balanced configuration $C_{ij} = -1/4$ for $i \neq j$. The residual list size is $|L_i| = 2^{0.1240d}$. The threshold is

$$C'_{12}(\alpha_0) = \frac{-1/4 + \cos^2(1.3131)/3}{\sin^2(1.3131)} \approx -0.244. \quad (30)$$

The solution probability is

$$p_{\text{sol}} = \frac{1}{2}(0.9405)^{d/2} = \frac{1}{2} \cdot 2^{-0.0443d}. \quad (31)$$

For the combined list $N_1 = 2|L_i|$ with $n_1 = 0.1240$, we have $\log_{N_1} p_{\text{sol}} \approx -0.360$. We optimise the QRW parameters. The optimal choice in the plateau region is

$$c = 0.3, \quad c_1 = 0.13, \quad c_2 = 0, \quad \rho = 0. \quad (32)$$

With these values, we compute the collision parameter carefully. We have

$$\log_2 \sin(\beta) = (c_2 - c_1) \cdot n_1 = -0.13 \cdot 0.1229 = -0.0160, \quad (33)$$

$$\cos^2(\beta) \approx 0.0219, \quad (34)$$

$$1 + \cos(\theta') = 1 - 0.244 = 0.756, \quad (35)$$

$$\log_2 \mathcal{W}_d(\beta, \theta')/d = \frac{1}{2} \log_2 \left(1 - \frac{2\cos^2(\beta)}{1+\cos(\theta')} \right) = -0.0431, \quad (36)$$

$$\rho_0 = \frac{-0.0160 - (-0.0431)}{0.1229} \approx 0.2205. \quad (37)$$

The marked fraction is therefore

$$\epsilon = N_1^{2c_1+\zeta-2c+\rho-\rho_0} = N_1^{0.26-0.360-0.2205} = N_1^{-0.3205}. \quad (38)$$

The remaining quantities are

$$\zeta = 2(0.3) - 0.360 = 0.240, \quad (39)$$

$$T_1 = N_1^{0.13} + N_1^{0.1603}(N_1^{0.065} + 1) \approx N_1^{0.2253}, \quad (40)$$

$$\text{FAS}_1 = N_1^{0.240} \cdot N_1^{0.2253} = N_1^{0.4653}, \quad (41)$$

$$\text{NbRep}_{\text{QRW}} = N_1^{0.3-0.240} = N_1^{0.060}, \quad (42)$$

$$T_{\text{QRW}} = N_1^{0.060}(N_1 + N_1^{0.7} \cdot N_1^{0.4653}) \approx N_1^{1.2253}. \quad (43)$$

Converting $T_{12} = T_{34} = 2^{0.1506d+o(d)} \approx 2^{0.151d}$.

5.4 Complexity of the Third Inner Call

After the first two inner calls, we have obtained the lists L_{12} and L_{34} of pairs of residual vectors that satisfy the firstlevel configuration constraints $C'_{12}(\alpha_0)$ and $C'_{34}(\alpha_0)$, respectively. The third inner call consists in finding all pairs

$$(\mathbf{z}_{12}, \mathbf{z}_{34}) \in L_{12} \times L_{34}$$

such that

$$\langle \mathbf{z}_{12} | \mathbf{z}_{34} \rangle \leq C_{\text{eff}}, \quad C_{\text{eff}} := \frac{\sin^2(\alpha_0)}{2R^2} - 1, \quad R^2 := 2 + 2C_{12}, \quad (44)$$

where α_0 is the outer prefiltering angle and C_{12} is the outer configuration entry. This condition is exactly the one given in [2, Lemma 6] and guarantees that the four residual vectors form a reducing tuple.

Probability of a solution. For two random vectors $\mathbf{z}_{12}$ and $\mathbf{z}_{34}$ of norm R , the probability that their inner product is at most C_{eff} is

$$p_{\text{sol}} = (1 - C_{\text{eff}}^2)^{d/2}. \quad (45)$$

There is no factor $1/2$ here, because the two lists L_{12} and L_{34} are distinct and the search is over ordered pairs $(\mathbf{z}_{12}, \mathbf{z}_{34})$.

The expected number of solutions per tuplefilter is therefore

$$|\text{Sol}| = |L_{12}|^2 p_{\text{sol}} = |L_{12}|^2 (1 - C_{\text{eff}}^2)^{d/2}. \quad (46)$$

Grover search. We use Grover's algorithm to find all solutions in $L_{12} \times L_{34}$. The search space has size $|L_{12}|^2$. The number of Grover iterations to find one solution is $O(1/\sqrt{p_{\text{sol}}})$, and to find all $|\text{Sol}|$ solutions we use amplitude amplification, which requires

$$T_{1234} = |L_{12}|^2 (1 - C_{\text{eff}}^2)^{d/4} = |L_{12}|^2 \sqrt{p_{\text{sol}}} \quad (47)$$

elementary operations.

Numerical values for the minimalmemory configuration. We take the minimalmemory parameters of [2, Proposition 17]:

$$|L| = 2^{0.1724d}, \quad \alpha_0 \approx 1.3131 \text{ rad}, \quad C_{ij} = -\frac{1}{4} \ (i \neq j).$$

Then

$$\sin^2(\alpha_0) \approx 0.9349, \quad R^2 = 1.5, \quad C_{\text{eff}} \approx -0.6884.$$

The firstlevel threshold is

$$C'_{12}(\alpha_0) = \frac{-\frac{1}{4} + \frac{\cos^2(\alpha_0)}{3}}{\sin^2(\alpha_0)} \approx -0.2443.$$

The residual list size is

$$|R_1| = |L| \mathcal{V}_d(\alpha_0) = 2^{0.1724d} \cdot 2^{-0.0486d} = 2^{0.1238d},$$

so that

$$|L_{12}| = |R_1|^2 (1 - C'_{12}(\alpha_0)^2)^{d/2} = 2^{0.2033d}.$$

The solution probability is

$$p_{\text{sol}} = (1 - C_{\text{eff}}^2)^{d/2} = 2^{-0.4635d},$$

hence

$$T_{1234} = 2^{0.2033d \cdot 2} \cdot 2^{-0.4635d/2} = 2^{0.2033d + o(d)}.$$

From Section 5.3, the first two inner calls cost

$$T_{12} = T_{34} \approx 2^{0.1517d+o(d)}.$$

Thus the total time per tuplefilter is dominated by the third call:

$$T_{\text{filter}} = T_{12} + T_{34} + T_{1234} \approx 2^{0.2033d+o(d)}. \quad (48)$$

Finally, we multiply by the number of tuplefilters and the number of repetitions of the outer loop. For the minimalmemory configuration, [2, Proposition 17] gives

$$\text{NbFilters} = 2^{0.0486d}, \quad \text{NbRep} = 2^{0.1069d}.$$

Therefore the total running time of the hybrid quantum 4sieve is

$$T_{\text{hybrid}} = \text{NbRep} \cdot \text{NbFilters} \cdot T_{\text{filter}} = 2^{0.1069d} \cdot 2^{0.0486d} \cdot 2^{0.2033d} = 2^{0.3588d+o(d)}. \quad (49)$$

We round this to

$T_{\text{hybrid}} \approx 2^{0.359d+o(d)}.$

6 Comparison

We now compare the hybrid quantum 4sieve with the classical 4sieve of [2] at the same classical memory. The relevant data for the minimalmemory regime is summarized in Table 1.

Algorithm	Classical memory	Running time
Classical 4sieve [2]	$2^{0.1724d}$	$2^{0.418d}$
Hybrid quantum 4sieve (ours)	$2^{0.1724d}$	$2^{0.359d}$
Grover 4sieve [2]	$2^{0.1724d}$	$2^{0.3276d}$

Table 1: Comparison at minimal classical memory.

At the minimal classical memory $M = 2^{0.1724d}$, the classical 4sieve runs in time $2^{0.418d+o(d)}$, whereas our hybrid algorithm runs in time $2^{0.359d+o(d)}$. The improvement factor is

$$2^{0.418d-0.359d} = 2^{0.059d},$$

which is an exponential speedup. For comparison, the pure quantum 4sieve of [2] achieves $2^{0.3276d+o(d)}$ at the same memory, so our hybrid is only slightly slower ($2^{0.0312d}$) than the best known quantum 4sieve.

The advantage of the hybrid is most pronounced in the lowmemory regime. As the allowed memory increases, the classical 4sieve improves rapidly: at $M = 2^{0.1887d}$ it runs in $2^{0.324d}$, and at $M = 2^{0.2075d}$ it reaches the classical 2sieve time $2^{0.2925d}$. Our hybrid, optimized for the minimalmemory point, does not immediately benefit from additional memory without a reoptimization of the QRW parameters. A full timememory tradeoff curve for the hybrid would require running the parameter optimization for several values of α_0 and C , which we leave for future work.

In summary, the hybrid quantum 4sieve demonstrates that quantum random walk techniques can be successfully integrated into the k -sieve framework. While it does not surpass the best pure quantum 4sieve, it offers a significant quantum advantage over the classical 4sieve in the regime where memory is severely constrained.

References

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